



Ford plans to go online

Ford has announced a plan to restructure its dealership model to sell EVs online-only. <https://digit.in/jun22-87>



Foxconn attacked in Mexico

Foxconn confirms their Mexico factory has been attacked with a ransomware. <https://digit.in/jun22-88>

AMD Ryzen 5 4500

Budget-friendly alternative

AMD's recent refresh for the Ryzen 5000 series processors was also accompanied by a couple of new processors under the Ryzen 4000 series as well. Yes, AMD is still releasing 'Zen 2' processors which are more appropriate for the entry-level segment. Since the Ryzen 5000 series processors didn't have any entry-level processors, users were left with older Ryzen 3000 series processors until now and that too were in short supply after a while. The introduction of these new Ryzen 4000 processors brings hope to the voluminous entry-level segment in India.

SPECIFICATIONS

The AMD Ryzen 5 4500 is a 6-Core / 12-Thread processor with similar specifications as its newer sibling, the AMD Ryzen 5 5500. You'd think that the presence of a 'Zen 3' successor would invalidate the Raison d'être of the Ryzen 5 4500 but the simple fact is that the INR 3K price difference between the two SKUs is too large to be ignored. Also, there's another major difference between the two processors in terms of the L3 cache being half of what the Ryzen 5 5500 comes with. This has a performance impact on several benchmarks but what needs to be seen is if the performance is in line with the pricing of the two SKUs.

Another question one might ask is how AMD has managed to release an older architecture processor which shares the same 7nm process node as their new 'Zen 3' processors. Wouldn't it have been better to make more Ryzen 5 5500 processors instead? The answer lies in how processors are manufactured and how each silicon wafer onto which the processor schematics are imprinted, have imperfections. These imperfections can reduce performance or completely disable certain modules within a processor. So a Ryzen 5 5600X wafer could yield Ryzen 5 5600X,



5600, 5500, etc. depending on how many imperfections there are.

The Ryzen 5 4500 is a 180mm2 monolith die which is the same as the Ryzen 4000 Mobile series processors. So this is basically a 'Renoir' mobile processor with the IGP turned off. The base clock is 3.6 GHz and the peak clock is 4.1 GHz. A huge chunk of the SKUs in the market are 'OEM' chips which don't come with a cooling fan but the retail units might come with the AMD Wraith Stealth CPU cooler. Being a 'Zen 2' processor, it will be supported by all boards based on the A320, A520, B350, B450, B550, X370, X470 and X570 motherboards. Only thing to remember is that there's no PCIe 4.0 lanes coming from the processor so investing in a PCIe 4.0 board might not be prudent.

PERFORMANCE

Our CPU benchmark tests begin with Cinebench and in the R20 single-threaded benchmark, we see that the Ryzen 5 4500 scores 478 points which is nearly a hundred points behind the Ryzen 5 5600. The multi-threaded run is about 400 points behind as well. In Cinebench R23, the performance gap is the same which works out to 16.5 per

cent in the single-threaded run and 9.4 per cent in the multi-threaded run. In Blender, the performance difference is an average of 9.8 per cent. In compression and decompression benchmarks, 7-Zip shows a different of up to 28 per cent during the compression benchmark but WinRAR is a lot worse showing a difference of 53 per cent.

An interesting thing that we'd noticed is that the memory and L3 Cache benchmarks favour the Ryzen 5 4500 over the Ryzen 5 5600. This is because of the 4500 being a monolith with a shared L3 across all cores whereas the Ryzen 5 5600 has two CCDs (Core Complex Dies) with their own L3 caches. As for L1 and L2 Cache benchmarks, the newer Ryzen 5

5600 shows a lead of approximately 5 per cent. Coming to gaming benchmarks, the Ryzen 5 4500 isn't behind the Ryzen 5 5600 by a huge margin. In Assassin's Creed Valhalla, the 5600 is 16-18 FPS ahead at 1080p but barely 3-4 FPS

ahead at 4K resolution. However, in games that have a more eSports focus such as Rainbow Six Siege, then the performance difference at 1080p can be quite high. Sometimes, going as high as 70-80 per cent.

VERDICT

Considering that you stand to save INR 3K over the Ryzen 5 5500, the new AMD Ryzen 5 4500 is a great buy. Day-to-day productivity tasks run just as fine and in some benchmarks, there's even a benefit thanks to its monolithic die. Only a few benchmarks show a drastic performance difference. So, if you're looking to a budget-friendly hexa core processor, then the Ryzen 5 4500 is a great buy.

—Mithun Mohandas

SPECIFICATIONS

CORES: 6 | THREADS: 12 | GPU CORES: NA | BASE CLOCK: 3.6 GHz | BOOST CLOCK: 4.1 GHz | PROCESS NODE: 7nm | L3 CACHE: 8 MB | L2 CACHE: 512 KB per core | TDP: 65 W.

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